

The Rank Function

Let $M = (E, \mathcal{I})$ be a matroid.

Recall that $\forall U \subseteq E$

$\text{rank}(U) = \text{size of a base of } U$

(well-defined b/c of the extension property
of $\mathcal{I}$).

Span

$$\text{span}(U) = \{i \in E : \text{rank}(U+i) = \text{rank}(U)\}$$

Immediately we have:

- $\forall U \subseteq E$, a base of U is a base of $\text{span}(U)$

- $\forall U \subseteq S \subseteq E$, B is a base of U , then $i \in \text{span}(S)$ iff $B+i$ & $\tilde{L}$

- $\forall T_1 \subseteq T_2$, $\text{span}(T_1) \subseteq \text{span}(T_2)$

Rank Function Characterization

The rank function satisfies:

- $\forall T_1 \subseteq T_2 : \text{rk}(T_1) \leq \text{rk}(T_2)$ (monotonicity)

- $\forall T_1 \in T_2, i \in E \setminus T_2$

$$\text{rk}(T_2 + i) - \text{rk}(T_2) \leq \text{rk}(T_1 + i) - \text{rk}(T_1)$$

(Non-increasing marginal value, aka diminishing returns)

pf: • A base of T_1 is an ind. set of T_2
s. 1st property is immediate.

- $\text{rk}(T_1 + i) - \text{rk}(T_1) = \begin{cases} 1 & \text{if } i \notin \text{Span}(T_1) \\ 0 & \text{if } i \in \text{Span}(T_1) \end{cases}$



But $\text{Span}(T_1) \subseteq \text{Span}(T_2)$.

Submodular Functions

A fn: $f: 2^E \rightarrow \mathbb{R}$ is called

submodular if: $\forall A, B \subseteq E$

$$\textcircled{*} \quad f(A \cap B) + f(A \cup B) \leq f(A) + f(B).$$

Lemma: A function $f: 2^E \rightarrow \mathbb{R}$ is submodular

iff it has non-increasing marginal values:

$$f(T_2 + i) - f(T_2) \leq f(T_1 + i) - f(T_1), \quad \forall T_1 \subseteq T_2$$

pf: One direction is immediate.

If f is submodular; $\forall T_1 \subseteq T_2$, and $i \notin T_2$

$$A = T_1 + i, \quad B = T_2$$

$$\begin{aligned} f(T_2 + i) + f(T_1) &= f(A \cup B) + f(A \cap B) \leq f(A) + f(B) \\ &= f(T_1 + i) + f(T_2) \end{aligned}$$

Proof — continued

For the other direction: consider any
 $A, B \subseteq E$. If $B \subseteq A$ — follows.

Suppose then: $B \setminus A = \{b_1, \dots, b_k\}$

$$\underline{f(A \cup B) - f(A)} = \sum_{i=1}^k f(A + b_1 + \dots + b_i) - f(A + b_1 + \dots + b_{i-1})$$

$$\leq \sum_{i=1}^k f(A \cap B + b_1 + \dots + b_i) - f(A \cap B + b_1 + \dots + b_{i-1})$$

$$= \underline{f(B) - f(A \cap B)}$$



Rank functions and submodularity

Lemma : $r : 2^E \rightarrow \mathbb{R}$ is the

rank function of a matroid $\mathcal{M} = (E, \mathcal{I})$

iff

1. $r(\emptyset) = 0$, $r(S+i) - r(S) \in \{0, 1\}$
2. r is submodular.

pf idea : To show, we need to find $\mathcal{M}$.

Let $\mathcal{I} = \{A \subseteq E : r(A) = |A|\}$.